

ZACATECAS INTL, MexicoWMO: **765255**Lat: **22.897N**Lon: **102.687W**Elev: **7141**StdP: **11.28**Time Zone: **-6.00 (MEX)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.7	31.6	3.0	8.3	42.2	6.5	10.0	46.1	33.7	63.5	29.8	61.2	3.9	180	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	31.5	87.5	55.4	84.4	54.4	82.6	54.1	61.1	73.0	60.3	72.6	59.7	72.4	13.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.3	91.9	65.0	55.8	86.9	64.5	55.4	85.4	64.4	30.9	73.3	30.2	73.1	29.6	72.3	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.1	23.2	21.7	19.4	90.1	5.4	2.0	15.5	91.6	12.4	92.8	9.3	93.9	5.4	95.4		
			16.1	64.4	4.0	2.8	13.2	66.4	10.9	68.0	8.6	69.6	5.7	71.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.6	51.4	54.9	58.0	63.4	67.5	68.2	65.8	65.5	63.8	60.5	55.5	52.0
	DBStd	6.82	4.10	4.37	4.29	3.71	3.67	3.10	2.49	2.37	2.55	3.59	4.45	4.31
	HDD50	84	34	11	3	0	0	0	0	0	0	0	9	27
	HDD65	1939	422	282	217	72	18	8	21	22	49	142	284	402
	CDD50	3945	77	150	252	402	541	546	488	481	415	327	175	91
	CDD65	323	0	0	1	24	94	104	44	38	15	3	0	0
	CDH74	5923	8	99	364	1114	1745	1197	477	442	199	201	67	10
	CDH80	1213	0	3	32	233	531	318	41	36	10	8	1	0
Wind	WSAvg	8.8	9.1	10.5	11.0	11.0	9.8	8.8	8.1	7.2	6.6	7.1	7.8	9.0
Precipitation	PrecAvg	17.5	0.6	0.5	0.2	0.2	0.6	2.8	3.7	3.4	3.6	1.2	0.4	0.2
	PrecMax	26.3	5.5	2.9	2.6	1.3	2.0	7.6	11.8	7.1	6.6	3.5	1.7	1.8
	PrecMin	7.9	0.0	0.0	0.0	0.0	0.0	0.7	0.8	0.8	1.0	0.0	0.0	0.0
	PrecStd	5.2	1.0	0.7	0.6	0.4	0.4	1.9	2.6	1.6	1.4	0.9	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.3	80.4	82.8	87.7	90.0	89.4	82.8	82.7	81.0	80.9	78.9	75.3
		MCWB	49.3	50.7	51.9	54.6	56.0	56.1	56.4	57.4	57.2	55.4	53.2	50.6
	2%	DB	73.0	77.2	80.5	84.4	87.9	86.2	80.9	80.7	78.8	78.7	75.5	73.1
		MCWB	48.4	49.7	50.6	53.2	55.4	55.7	56.5	57.4	57.3	55.2	52.4	49.3
	5%	DB	70.1	75.0	78.6	82.6	85.1	83.9	79.0	78.9	75.6	75.6	73.6	71.3
		MCWB	47.3	49.0	50.1	52.4	54.2	55.7	56.8	57.7	57.7	55.2	51.7	48.8
	10%	DB	67.9	71.9	75.4	80.7	82.7	80.9	76.9	76.4	73.7	73.7	71.6	68.3
		MCWB	46.4	48.2	49.0	51.3	53.5	56.0	56.9	57.3	58.1	55.1	50.9	47.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	53.7	54.0	56.0	59.5	61.8	61.2	62.0	62.7	61.2	58.1	54.1
		MCDB	68.9	73.0	75.0	80.8	79.0	75.8	73.0	73.1	72.8	72.2	71.9	67.1
	2%	WB	50.6	51.6	52.4	54.5	57.5	60.2	60.1	60.8	61.1	59.8	56.1	51.9
		MCDB	67.4	72.4	74.9	80.5	78.2	74.7	72.6	72.9	71.9	71.8	70.1	68.2
	5%	WB	49.3	50.4	51.1	53.3	56.4	59.2	59.2	59.9	60.2	58.3	54.4	50.4
		MCDB	66.0	71.4	74.4	79.8	78.1	74.4	72.5	72.6	71.1	70.9	68.0	67.0
	10%	WB	47.7	49.1	50.0	52.2	55.3	58.2	58.3	58.9	59.4	57.1	52.7	49.1
		MCDB	64.1	69.7	73.5	78.2	77.9	73.4	71.5	72.0	70.2	69.4	66.7	65.1
Mean Daily Temperature Range	5% DB	MDBR	32.5	33.2	34.6	34.9	31.5	25.0	22.2	22.8	21.8	27.7	32.0	32.4
		MCDBR	36.9	38.2	38.2	37.0	34.6	30.7	26.3	26.9	25.9	31.8	35.0	36.7
		MCWBR	18.9	18.6	17.9	16.3	13.8	10.9	9.4	10.1	11.0	14.5	17.0	18.7
	5% WB	MCDBR	30.3	33.0	33.9	34.7	28.9	22.6	20.8	21.5	20.0	24.6	27.6	31.1
		MCWBR	16.1	16.3	16.5	15.6	10.9	8.4	8.6	9.1	9.4	11.9	14.0	16.4
Clear-Sky Solar Irradiance	taub	0.265	0.276	0.278	0.299	0.354	0.348	0.352	0.344	0.345	0.310	0.274	0.272	
	taud	2.550	2.508	2.503	2.422	2.325	2.381	2.411	2.445	2.435	2.524	2.594	2.568	
	Ebn at Noon	322	324	326	318	297	296	295	299	299	307	317	316	
	Edh at Noon	29	32	34	38	41	39	38	36	36	31	28	28	
All-Sky Solar Radiation	RadAvg	1483	1796	2140	2341	2318	2129	2037	2009	1758	1738	1573	1407	
	RadStd	106	156	109	132	121	139	145	108	127	111	120	90	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	+0.63	N/A	N/A	N/A	+0.61	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+0.77	N/A	N/A	N/A	+0.61	N/A	N/A	-202	+356	N/A

Nomenclature: See separate page